

Shangmin Chen

347-322-4133 | shangminch@gmail.com | linkedin.com/in/shangmin-chen | github.com/shangmin-chen

EDUCATION

Boston University

Bachelor of Arts in Computer Science

Boston, MA

September 2022 – May 2026

EXPERIENCE

Software Engineer

September 2024 – December 2024

BU Spark!

Boston, MA

- Developed full-stack food waste platform using Django, JavaScript, and SQLite, connecting 500+ students and reducing campus waste by 10%
- Built RESTful API with sub-100ms query times and deployed via CI/CD pipelines on Heroku
- Implemented Auth0 authentication and Django security middleware, ensuring zero security incidents
- Accelerated release by 1 week using Agile methodology in team of 5, contributing to 7 iterations

IT Consultant

January 2024 – July 2025

Boston University Engineering IT

Boston, MA

- Automated software deployment for 100+ devices using Microsoft Deployment Toolkit and created custom USB deployment solution for Windows 11, reducing deployment time by 40%
- Managed 200+ user accounts and permissions via NAS, conducting access audits and boosting system uptime by 15%
- Enhanced endpoint security with CrowdStrike for 50+ devices, resolving 20+ tickets/week with 100% incident resolution rate

PROJECTS

Crime Analytics & Forecasting Platform | *Python, Streamlit, Prophet, Scikit-learn, GeoPandas* December 2024

- Built crime analytics platform processing 500K+ Boston crime records with Prophet time-series forecasting and DBSCAN clustering for hotspot identification across 12+ police districts
- Implemented advanced data pipeline with seasonal decomposition and confidence intervals, generating 2-month crime predictions with train/test validation
- Developed interactive dashboard using Streamlit and Plotly, creating 121 crime-type heat maps with Folium and real-time geospatial visualizations

Audio Transcription Platform | *Java Spring Boot, Python FastAPI, React, PostgreSQL* September 2025

- Built microservices platform with Spring Boot API, Python FastAPI service, and React frontend, implementing async audio processing with OpenAI Whisper and real-time status polling
- Designed PostgreSQL database with 4 related tables, Flyway migrations, and optimized indexing, supporting multiple audio formats with comprehensive error handling
- Deployed containerized application using Docker Compose with inter-service communication, CORS configuration, and health monitoring endpoints

Decentralized ML Model Marketplace | *Solidity, Circom, IPFS, Express.js, TensorFlow* May 2024

- Built decentralized marketplace using Solidity smart contracts with ERC-20 tokens, implementing 6-state transaction management, automated candidate selection, and economic incentive mechanisms with stake slashing
- Integrated IPFS distributed storage with Ethereum blockchain using Thirdweb SDK, enabling secure model uploads with SHA256 hash verification and automated ML training workflows
- Implemented zero-knowledge proof system using Circom and Groth16 protocol for privacy-preserving accuracy verification, with MetaMask integration and real-time blockchain event monitoring

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, TypeScript, SQL (PostgreSQL), HTML/CSS

ML/AI Frameworks: TensorFlow, Scikit-learn, Prophet, OpenAI Whisper, Pandas, NumPy, Time-Series Analysis

ML Pipeline & Deployment: Model Training, Validation, Deployment, Microservices Integration, Real-Time Inference

Data Processing: Large Dataset Processing (500K+ records), ETL Pipelines, DBSCAN Clustering, Geospatial Analysis

Blockchain & ML Integration: Solidity, Smart Contracts, IPFS, Zero-Knowledge Proofs, Decentralized ML Model Management

Tools & Infrastructure: Docker, Docker Compose, Streamlit, Plotly, Folium, PostgreSQL, Health Monitoring